COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	8/07/2026	**DATE SUBMITTED:**	8/09/2026
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ACTIVITY NO. 3.2 Program Control

1. PROCEDURE OUTPUT

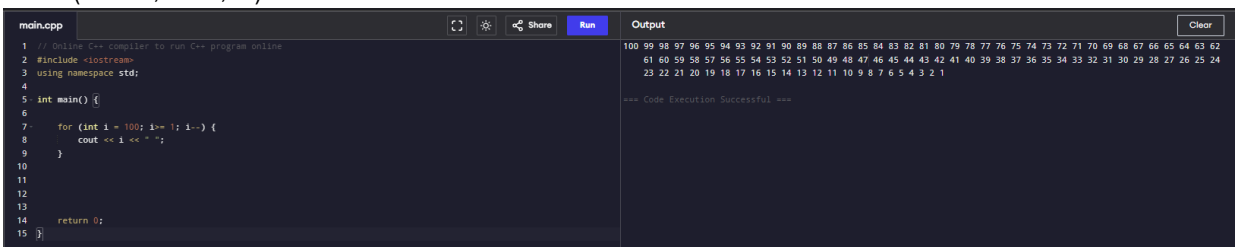
What procedures did you perform?

a. for (i = 1; i <= 10; i++)



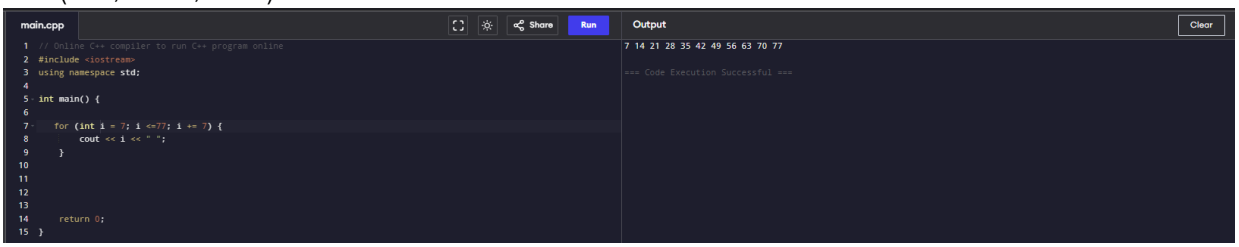
The program uses a for loop to print the numbers from 1 to 10. The variable i starts at 1, and the loop continues as long as i is less than or equal to 10. After each loop, i++ increases the value by 1. The cout statement then displays each number in order, resulting in the sequence from 1 to 10.

b. for (i = 100; i>= 1; i--)



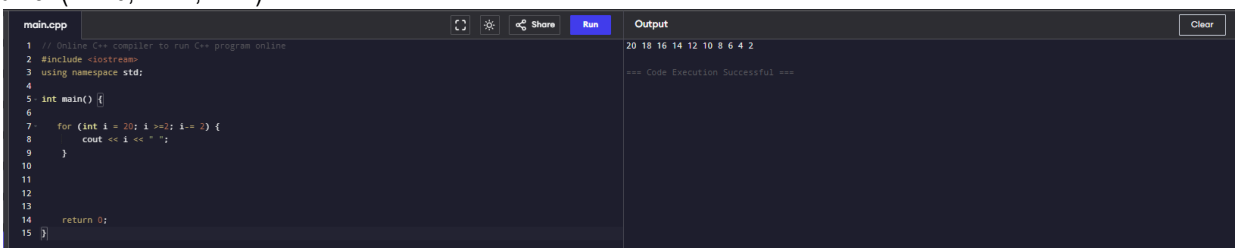
Here, the loop starts with i at 100 and moves backward until it reaches 1. The i-- part is responsible for subtracting 1 from the value every time the loop repeats. Since cout prints the current value of i, the numbers appear in descending order from 100 to 1.

c. for (i = 7; i <=77; i += 7)



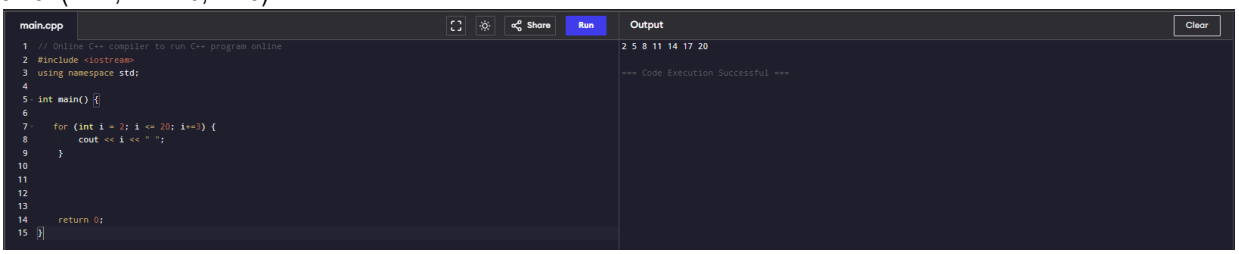
This loop is set to begin at 7 and increase by 7 on every repetition. The condition i <= 77 keeps the loop running until it reaches 77. Because i += 7 adds 7 each time, the program prints the multiples of 7 in order, starting from 7 and ending at 77.

d. for (i = 20; i >=2; i-= 2)



In this example, the loop begins with i at 20 and decreases by 2 each time through i -= 2. The condition i >= 2 makes sure the loop stops once the value goes below 2. This allows the program to display only the even numbers, starting from 20 and going down to 2.

e. for (i = 2; i <= 20; i+=3)



This loop starts at 2 and keeps increasing by 3 using i += 3. The condition i <= 20 limits the loop so it stops once the value goes beyond 20. As a result, the program prints the numbers that follow a difference of 3, starting from 2 up to 20.

f. for (i = 99; i >= 0; i -= 11)



The loop starts at 99 and decreases by 11 every time using i -= 11. It continues as long as i is greater than or equal to 0. Because the value goes down by 11 on each repetition, the program prints the sequence from 99 down to 0.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

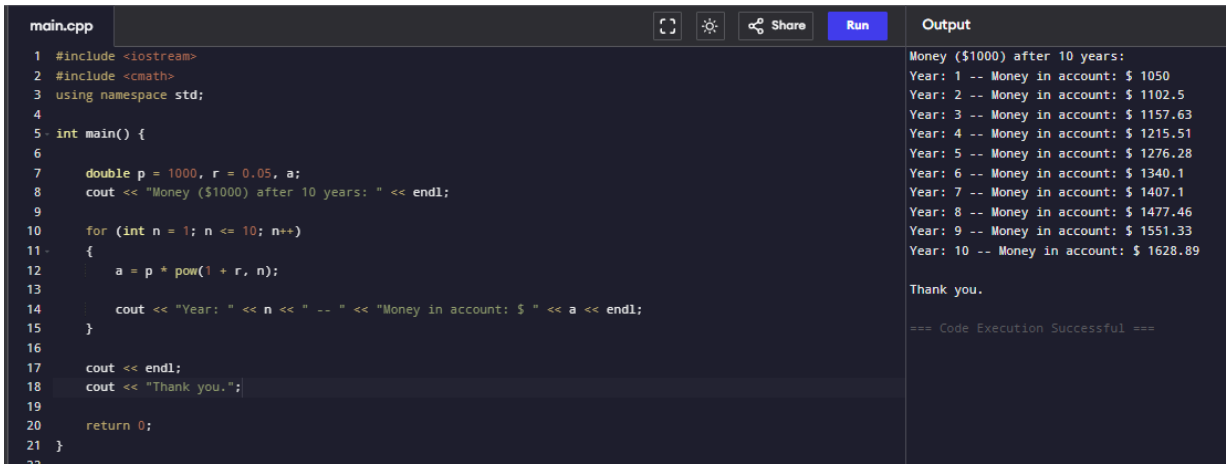
Copy and paste the questions from the lab and include your answers after.

1. Create a program that uses for structure to sum all the even integers from 2 to 100. Sum is 2550.



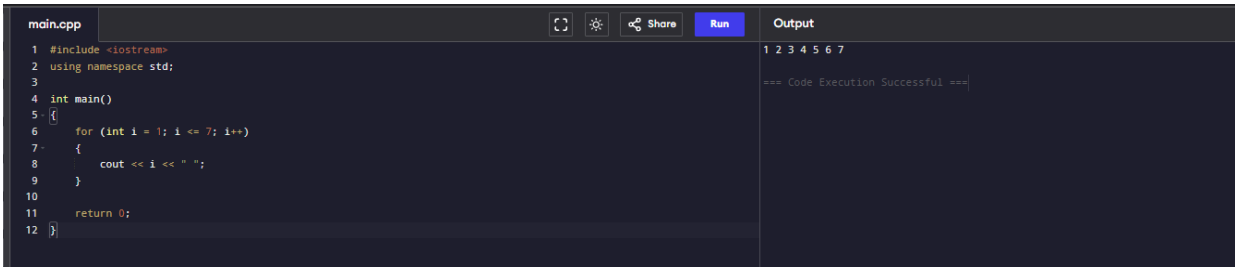
The program uses a for loop to go through the even numbers from 2 to 100. The variable i starts at 2 and increases by 2 each time, so only even numbers are included. Each value of i is added to sum, and after the loop finishes, the program displays the total, which is 2550.

2. Create a program that computes compound interest using the for structure. Consider the following problem statement: A person invests $1000.00 in a savings account yielding 5 percent interest. Assuming that all interest is left on deposit in the account, calculate and print the amount of money in the account at the end of each year for 10 years. Use the following formula for determining these amounts: a = p (1 + r) ^n; here p is the original amount invested (i.e. principal), r is the annual interest rate, n is the number of years, a is the amount on deposit at the end of the nth year



The program calculates the amount of money in the account for each year using the compound interest formula. The for loop starts at year 1 and continues until year 10. For every repetition, pow() is used to raise (1 + r) to the current year, then the result is multiplied by the initial investment. This gives the amount of money in the account at the end of each year.

3. Write for statements that print the following sequences of values:
1, 2, 3, 4, 5, 6, 7



The program uses a for loop that starts with i = 1 and runs until i reaches 7. After each repetition, i++ increases the value by 1. The cout statement then prints the current value of i, producing the sequence from 1 to 7.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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